



PYTHON ROADMAP

Step by Step to Become a Python Developer



Learn. Practice. Build. Automate.
Python for everyone and everywhere.

1



Python Basics

- Setup Python, IDE (VS Code / PyCharm)
- Syntax, Comments, Variables, Data Types
- Input / Output, Type Casting
- Operators, Expressions

⌚ 1–2 Weeks

OUTCOME

Write and run simple Python programs.



2



Control Flow & Functions

- if, elif, else
- for, while Loops
- break, continue, pass
- Functions, Parameters, Return Value

⌚ 1–2 Weeks

OUTCOME

Build logic using conditions, loops and functions.



3



Data Structures

- Lists, Tuples, Sets, Dictionaries
- Indexing, Slicing
- List/Dict Comprehensions
- Built-in Functions

⌚ 1–2 Weeks

OUTCOME

Work with Python collections efficiently.



4



OOP in Python

- Classes & Objects
- Constructors (__init__)
- Inheritance, Polymorphism
- Encapsulation, Abstraction

⌚ 2–3 Weeks

OUTCOME

Build reusable and maintainable code using OOP.



5



Modules & Packages

- Built-in Modules (math, datetime, os, sys)
- Creating & Importing Modules
- Packages and __init__.py
- pip and Virtual Environment

⌚ 1–2 Weeks

OUTCOME

Organize code using modules and packages.



6



File Handling & Exceptions

- Read / Write Files
- with open()
- Exception Handling (try, except, finally)
- Raising Custom Exceptions

⌚ 1–2 Weeks

OUTCOME

Handle files and errors like a pro.



7



Database Basics

- SQL Basics (SELECT, INSERT, UPDATE, DELETE)
- Working with SQLite
- Using Python (sqlite3 module)
- CRUD Operations

⌚ 1–2 Weeks

OUTCOME

Connect Python apps with databases.



8



Python for Web (Basics)

- HTTP & Requests (requests module)
- REST APIs – GET, POST, PUT, DELETE
- JSON Handling
- Introduction to Flask (Basics)

⌚ 2–3 Weeks

OUTCOME

Call APIs and build simple web apps.



9



Advanced Topics (Optional but Useful)

- Regular Expressions
- Lambda, Map, Filter, Reduce
- Decorators, Generators
- Multithreading / Multiprocessing (Basics)

⌚ 2–3 Weeks

OUTCOME

Write efficient and powerful Python code.



10



Real-World Projects

- Apply everything you learned
- Build 2–3 real-world projects
- Document and showcase your work
- Keep improving and learning

⌚ Ongoing

OUTCOME

Strong portfolio that gets you hired.



ESSENTIAL TOOLS



VS Code



PyCharm



Git & GitHub



Jupyter



pip

PROJECT IDEAS

✓ To-Do App (CLI / GUI)

✓ Weather App

✓ Blog with Flask

✓ Expense Tracker

✓ E-commerce API

✓ Data Analysis Dashboard

TIPS FOR SUCCESS

✓ Practice every day

✓ Solve problems

✓ Build projects

✓ Read documentation

✓ Learn by teaching others



Consistency is the key! Keep learning, keep building, and enjoy the journey.

